

BA 336

Information Technology

09/30/2026

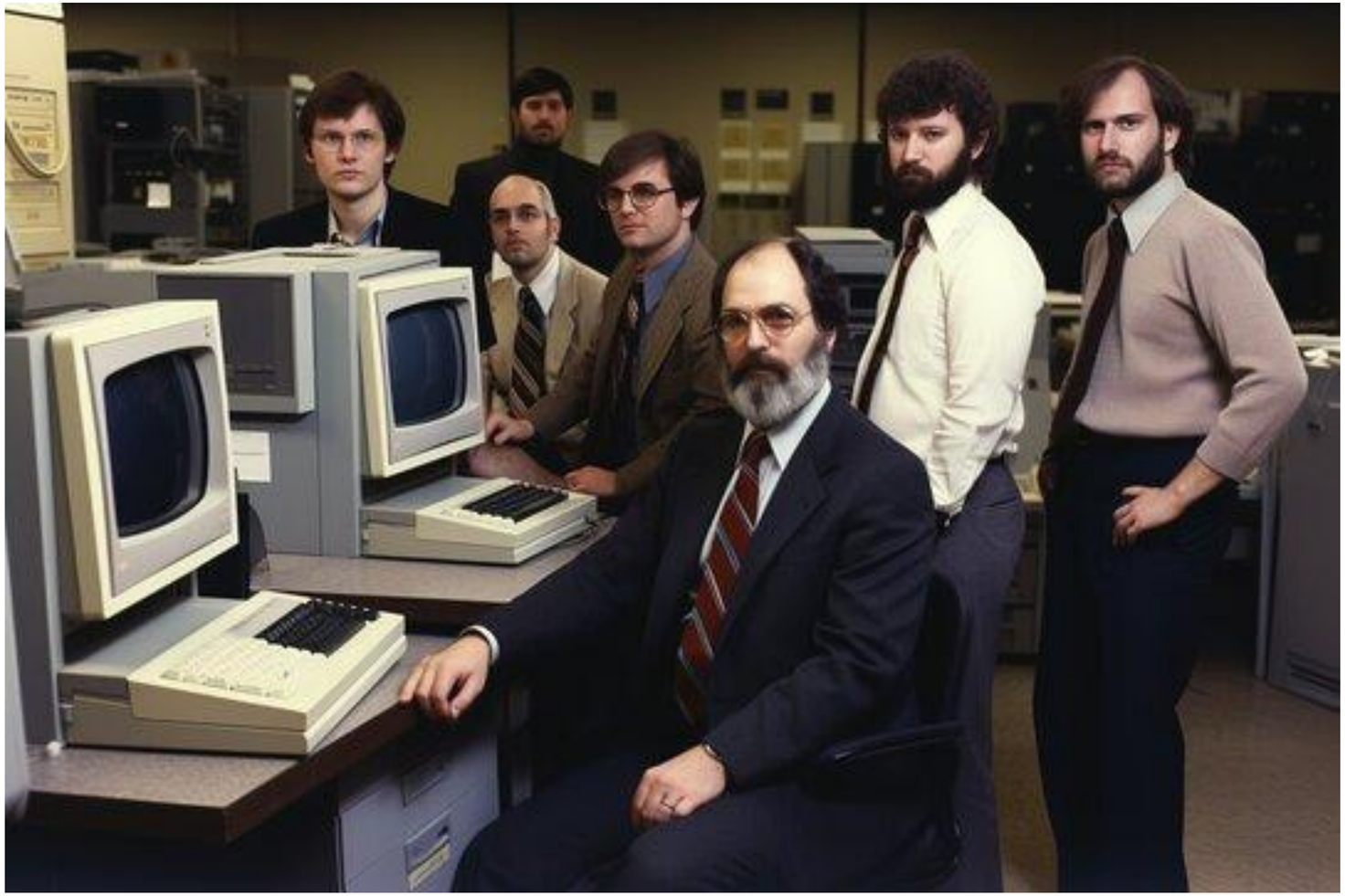
Shaw Millerman

The School of Business

PORTLAND STATE UNIVERSITY

WHY AM I REQUIRED TO BE HERE?

Reason #1: Meet the IT Team!



WHY AM I REQUIRED TO BE HERE?

Reason #2: Every Company is a Tech Company

Forged in Silicon Valley

• Modern business tools, software management frameworks, and data strategies weren't invented in traditional corporate boardrooms.

• They were pioneered, tested, and perfected inside Silicon Valley's tech ecosystem before scaling globally.

Escaped Containment

• Tech is no longer isolated to software companies. Software and data strategies now actively drive *every* industry on the planet.

• From global finance to local service providers—every enterprise is now a tech company, and every role is a tech role.

01

Digital Infrastructure

Core operations in every sector rely directly on cloud computing, enterprise software, and robust data pipelines.

02

Data-Driven Decisions

Competitive strategy across modern markets requires fluency in analytics, data literacy, and technological agility.

03

Universal Tech Roles

Regardless of your specific job title or discipline, you will interact with and manage technology every single day.

WHY AM I REQUIRED TO BE HERE?

Reason #3: Data Literacy Is Not Optional

Data is Messy

• Data doesn't emerge from software clean and ready for a dashboard. It is inherently messy, duplicated, badly formatted, and siloed.

• Navigating real-world business data requires technical confidence, strategic patience, and hands-on data literacy.

A Non-Negotiable Core Competency

• Data literacy is no longer a "nice to have" luxury or a niche skill reserved for specialized analytics teams.

• Understanding how data flows, transforms, and informs strategic decisions is now a fundamental requirement for modern business leadership.

The Data Lifecycle

01 Sourcing
Extracting raw, unstructured data directly from databases, software tools, and enterprise systems.

02 Cleaning
Resolving duplicates, handling missing values, and scrubbing errors from raw data streams.

03 Structuring
Organizing, transforming, and modeling clean data into standardized operational formats.

04 Reporting/Action
Translating structured datasets into visual dashboards, key metrics, and strategic business execution.

WHY AM I REQUIRED TO BE HERE?

3 Pillars Revolutionizing Business & Finance

01

Artificial Intelligence

• Impacting every operational layer and strategic decision-making process.

• Drives automation, predictive modeling, and intelligent insights across modern global enterprises.

02

Blockchain

• Redefining trust, transparency, and data integrity across the global financial industry.

• Enables decentralized ledgers, secure smart contracts, and verifiable transactional provenance.

03

Quantum Computing

• The horizon technology that completely rewrites modern computational limits.

• When combined with AI, unlocks exponential processing capabilities for complex problem solving.

On the Dangers of AI

01 Environmental Damage

- Massive power consumption standard across data centers.
- Cooling servers requires vast amounts of water heavily contaminated with heavy metals.

02 Cognitive Decline

- Brain scans demonstrate virtually zero active engagement during copy-and-paste tasks.
- Passive reliance inhibits actual learning and critical thinking processes.

03 Existential Risk

- Growing concerns surrounding long-term control and safety dynamics.
- Threat vectors surrounding unintended autonomous behaviors and robot deployment.

Agenda

- Welcome and Attendance
- Introductions
- The Syllabus, Canvas, Course Expectations
- Information Systems



About Me

- Travel, family, extra in film and TV
- BA English, WWU
- MBA, University of Oregon
- 15 years in Tech (13 years Operations, 6 in Data and Analytics
- Industry background vs. academic background









About You

- Your name
- Something fun about you
- Are you studying general business or specialization?
- What do you want to accomplish with a business degree?
- What were your honest thoughts about this class when you registered?

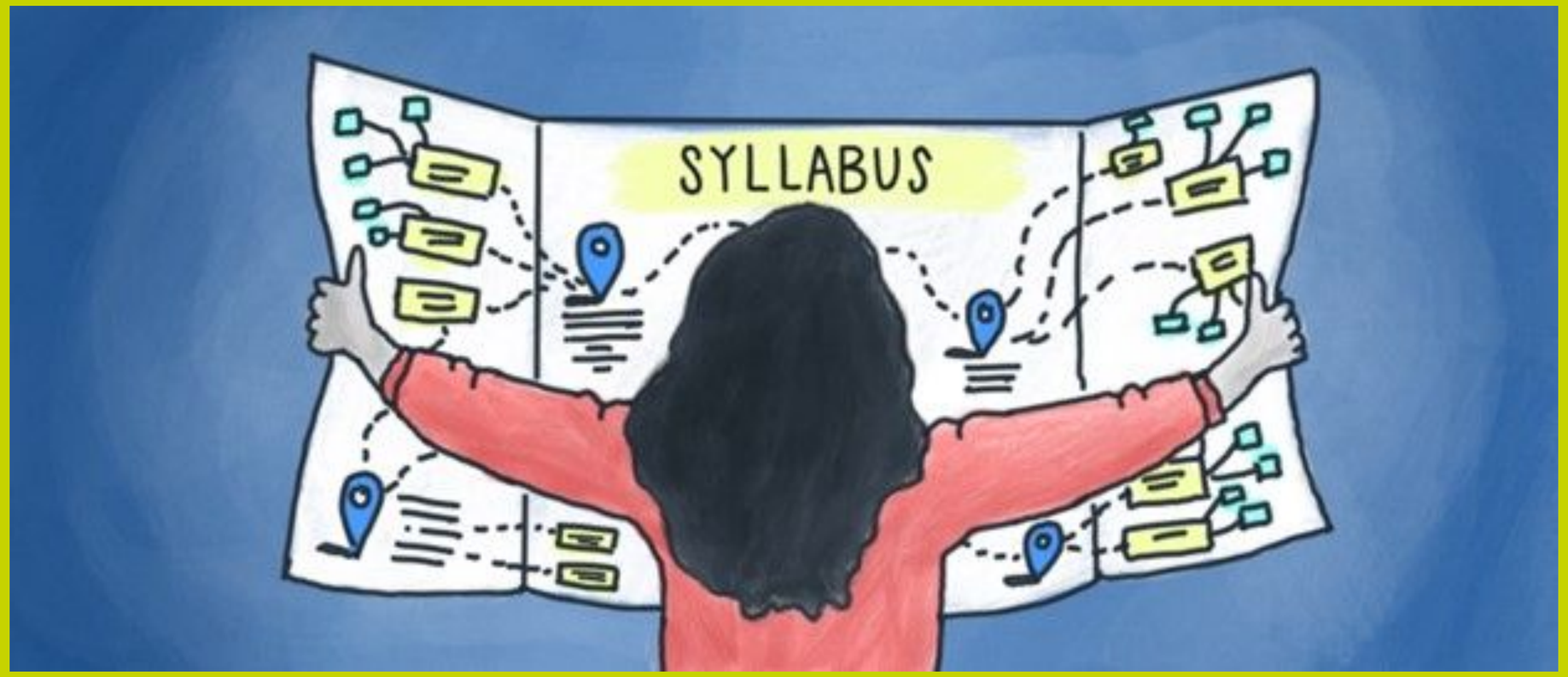


Agenda

- Introductions
- The Syllabus, Canvas, Course Expectations
- Tech and the Modern Enterprise
- Excel homework preview



The Canvas and Syllabus Tour



Course Logistics and Expectations



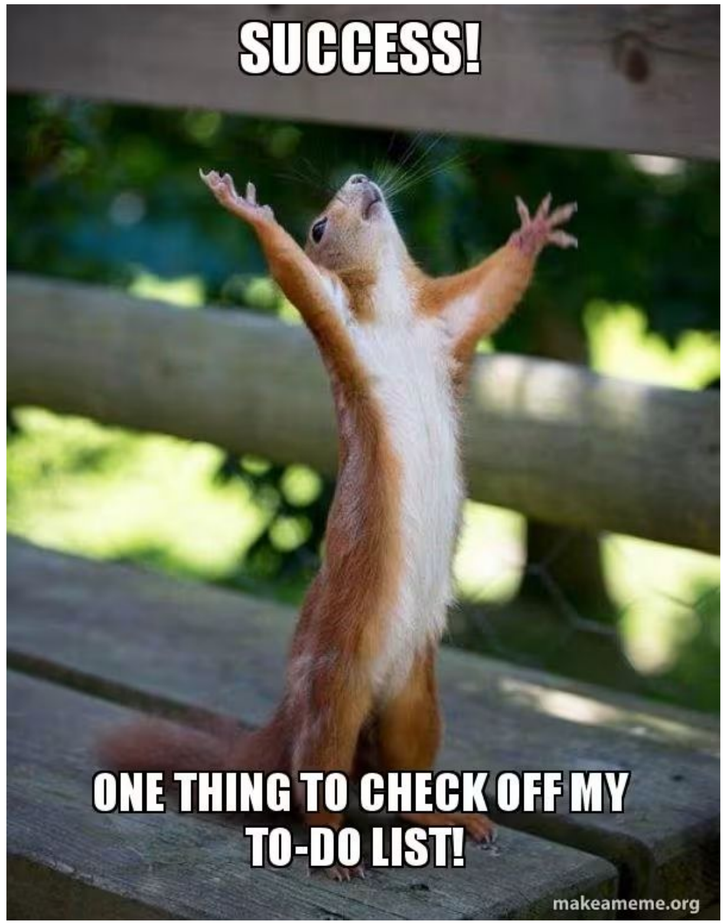
- Attendance is required. You have two "no-questions-asked" absences as grace.
  - Information delivered in class will make your life **EASIER and save you time**.
- In-Class Focus: **Laptops and phones closed**
- Late work policy: After due date, 12% deduction. Five days only.

Assignments and Assessments

Scored Items	Point Value Each	Total Point Value
Excel Activities • **6 Guided** • **6 Practice** • **6 Challenge**	1 point each 2 points each 4 points each	42 points
Comparative Analysis of Critical Terms • **Chap 1, 6, 10** • **Chap 16, 17, 18**	4 points each	8 points
Excel Projects	6 points each	12 points
Midterm	6 points	6 points
Google Sheets vs. Excel	4 points	4 points
Information System in the News	4 points	4 points
Prompt Engineering	8 points each	8 points
Case Study	8 points	8 points
Final Exam	8 points	8 points
Total		**100 Points**

How to Succeed in BA 336

1. Don't Get Behind: This is a hands-on, skills-based course. Pace yourself.
2. Move Slow, Retain, and Build: The skills build on each other.
3. Pay Attention to Details: Especially in Excel.
4. When You Need Help (Reach Out EARLY!)




Have a break,
have a KitKat.®

PORTLAND STATE UNIVERSITY • BA 336 ESSENTIALS OF INFORMATION TECHNOLOGY

Setting the Stage: Technology & the Modern Enterprise

Week 1 • Navigating promise, peril, and digital transformation through the manager's lens.

The New Business Reality: Asset-Less Giants

Technology has fundamentally rewritten market dominance without traditional physical footprints.

Uber

World's largest taxi service owns zero vehicles for hire.

Airbnb

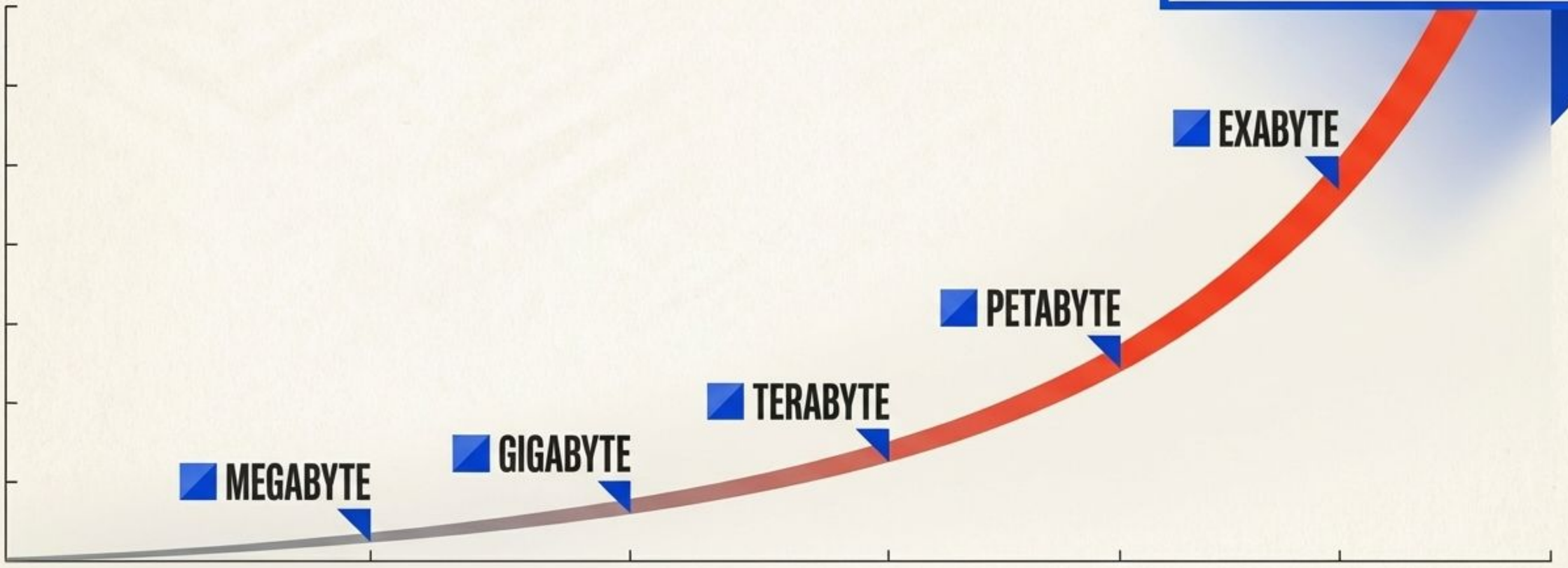
Largest accommodations provider owns no hotels or real estate.

Meta & Google

Dominant media destinations create zero original media content.

THE DAILY DATA FLOOD

0.5 ZETTABYTES
(A sextillion bytes. 21 zeros.
Generated every single day.)



Gathering, structuring, and making sense of this data
is the defining challenge of the modern enterprise.

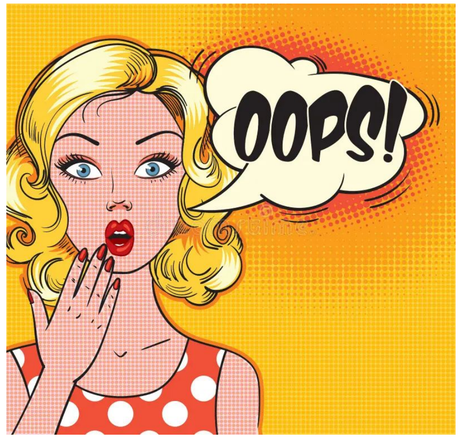
72%

OF FIRMS ARE FAILING

AT DIGITAL TRANSFORMATION

The Manager's Dilemma: Digital Transformation

While digital tailwinds accelerate global markets, organizational leadership struggles to keep pace.



72%

Firms Failing at Transformation

Managers often ill-equipped for rapid silicon-era shifts.

20%

Directly Digital Economy

With the remaining 80% running entirely on digital infrastructure.

The Trillion-Dollar Club & The AI Arms Race

Elite tech giants dominate global valuation and are heavily re-investing capital into artificial intelligence.

Apple & Microsoft

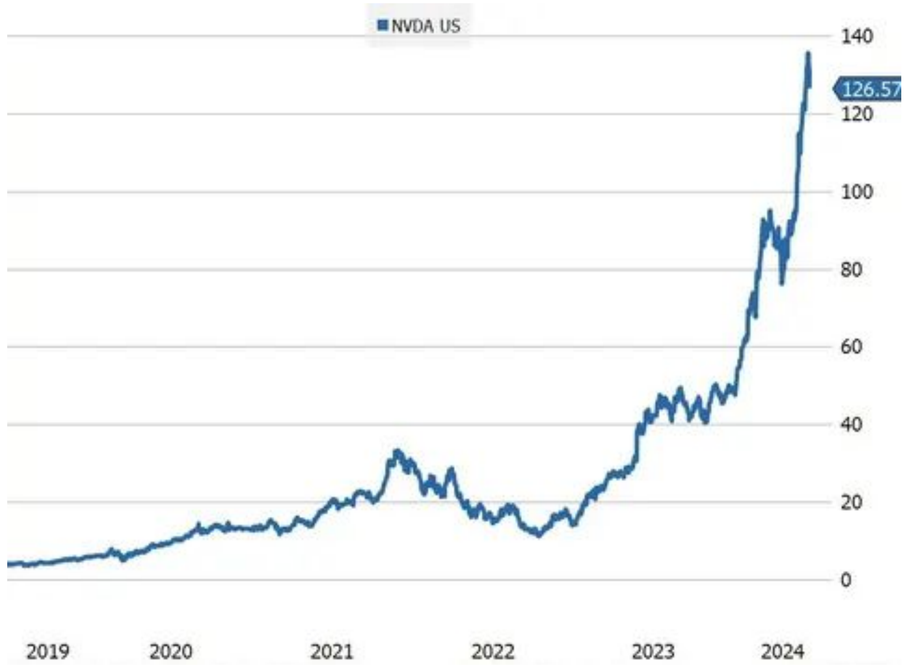
Platform ecosystem leaders staking massive claims in consumer and enterprise AI.

Alphabet & Meta

Pioneering neural networks, foundational models, and open-source AI infrastructure.

Nvidia

The indispensable microchip "arms merchant" commanding 80-95% of high-end AI hardware.



Artificial Intelligence: Promise & Peril

Unprecedented productivity gains paired with profound ethical, security, and societal risks.

⚡ Extraordinary Promise

Programming Speed: Instant code lookups and syntax support.

Workflow Automation: Eliminating tax & legal grunt work.

Life Sciences: Accelerating vaccine design and medical breakthroughs.

⚠ Severe Peril

Bias & Hallucinations: Fabricated data and algorithmic discrimination.

Security Vulnerabilities: Unvetted enterprise tool exposure.

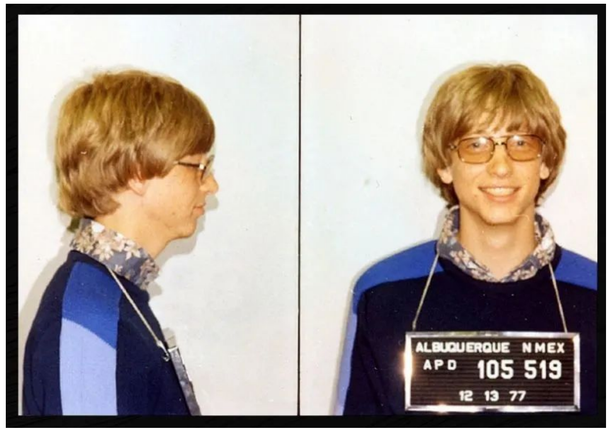
Resource Footprint: Massive server farm electricity and climate impact.

It's Your Revolution: Youth & Entrepreneurship

Many of the world's most transformative tech firms were founded by university students and twenty-somethings.

Microsoft & Apple

Bill Gates and Steve Jobs dropped out in their early twenties to redefine personal computing.



Meta & Google

Mark Zuckerberg launched Facebook at 19; Page & Brin started Google as Stanford PhD students.



Stripe & Canva

Patrick Collison and 19-year-old Melanie Perkins built multi-billion-dollar modern powerhouses.



Startup Superchargers

Modern tooling lowers barriers, but rigorous fundamentals separate builders from "wantrepreneurs."

Enabling Forces

Cloud Computing: Rent powerful infrastructure instantly.

App Stores: Global zero-cost code distribution.

Crowdfunding: Kickstarter funding without equity loss.



Tech Across the Enterprise: Finance & Accounting

Information technology is the operational lifeblood of financial valuation and audit accuracy.

Finance & Investment Banking

Tech startups drive IPO markets and global M&A activity.

Automated **algorithmic trading** handles 60% of U.S. equity volume.

Accounting & Governance

Audit **reliability** is directly tied to **underlying information systems**.

Accounting consulting divisions (e.g., Deloitte tech arms) now outgrow audit practices.

Tech Across the Enterprise: Marketing & Operations

Real-time consumer tracking and automated value chains redefine customer acquisition and production.

Marketing Transformation

Shift from legacy media to SEM, SEO, CRM, and influencer ecosystems.

CMOs now **spend more on technology** than any other functional department.

Operations & Supply Chain

Factory automation, Big Data analytics, and **IoT sensors optimize production.**

Agricultural drones cut crop survey costs from $1,000/hr to under $1,000 total.

Tech Across the Enterprise: HR & Corporate Law

Recruiting databases, professional social graphs, and automated compliance reshape organizational talent.

Human Resources

Automated **applicant tracking systems** screen résumés via keyword matching.

LinkedIn makes top talent vulnerable to competitor headhunting.

Corporate Law

Exponential rise in patent applications and intellectual property disputes.

Generative **AI automates legal document review** (with human oversight required).

Managing the Hype Cycle: Timing & Strategy

Technology is fashion-driven. Managers must assess timing between early adoption risks and lagging obsolescence.

1. Innovation Trigger

Early proofs-of-concept and VC funding spark initial excitement.

2. Inflated Expectations

Frenzied media hype peaks before tangible ROI materializes.

3. Trough of Disillusion

Interest craters as failures mount and projects are abandoned.

4. Slope of Enlightenment

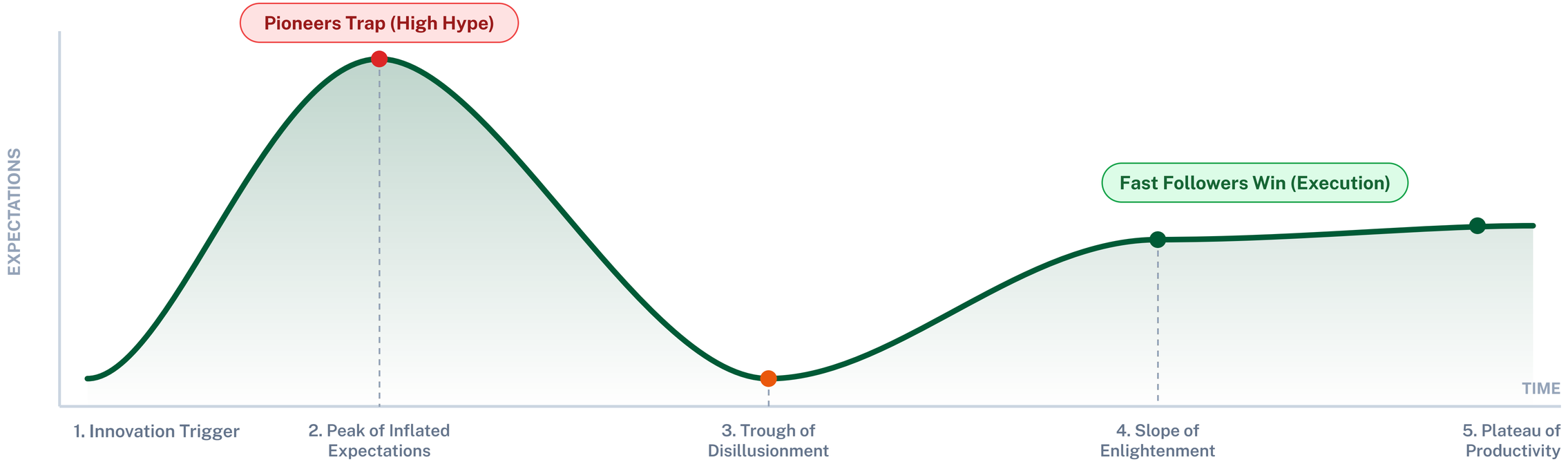
Practical business use cases emerge with proven benefits.

5. Plateau of Productivity

Mainstream commercial adoption and steady enterprise returns.

Hype Cycle Lessons: Pioneers vs. Fast Followers

History shows that brand invention doesn't guarantee market victory. Execution and timing dictate winners.



Xerox & Apple

Xerox invented the graphical user interface; Apple commercialized it.

IBM Watson & Cloud

IBM pioneered utility computing and Watson AI, but lost cloud leadership to Amazon.

Palm & iPhone

Blackberry and Palm dominated early mobile before Apple and Android disrupted the space.

BA 336 Course Roadmap: The Tech Industry Overview

A comprehensive look at the foundational pillars powering modern business enterprises.

Hardware & Software

Silicon microprocessors, operating systems, cloud architecture, and open source ecosystems.

Strategy & Networks

Competitive advantage, network effects, platform economics, and digital business models.

Data & AI Innovation

Databases, business intelligence, machine learning, and generative AI systems.

The Core Takeaway: Every Company is a Tech Company

Technology is merely the lens; organizational strategy and operational execution define modern success.



Multidisciplinary Lens

Engineering, healthcare, and liberal arts majors uniting to build tech-fluent business acumen.

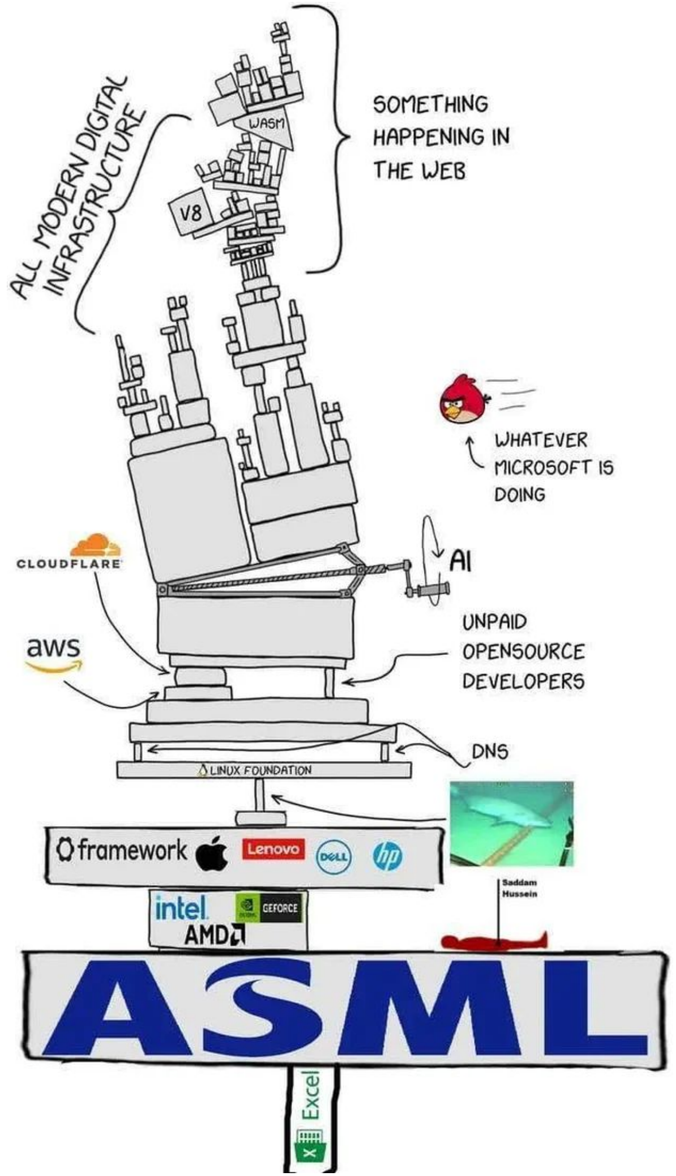


Managerial Playbook

Placing technology at the center of managerial decision-making for a resilient career.

Week 1

Why Excel?


ALL MODERN DIGITAL INFRASTRUCTURE
WASM
V8
SOMETHING HAPPENING IN THE WEB
WHATEVER MICROSOFT IS DOING
AI
CLOUDFLARE
aws
UNPAID OPENSOURCE DEVELOPERS
DNS
LINUX FOUNDATION
framework
Lenovo
DELL
hp
intel
AMD
GEFORCE
Saddam Hussein
ASML
Excel


Buh Bye

The 21st Century Challenge

Why Data Literacy is the Ultimate Career Differentiator.









Organization: The 21st Century Challenge

20th Century	21st Century
Building physical infrastructure (roads, power lines).	Organizing the digital world. Structuring data for ethical, useful application.

The Anatomy of Failure: Why 72% of Digital Transformations Collapse

Dissecting the systemic fractures across Leadership, Strategy, and Execution.

Pillar 1: Leadership Deficits (The Who)

Lack of Tech-Centric Leadership

Failing: The Non-Technologist

Attempting to lead tech initiatives without tech vision (e.g., AOL-Time Warner).



Winning: The Tech-Centric Visionary

Tech-fluent leadership at the helm driving the narrative.



Managers are Ill-Equipped

Failing: The Preparedness Gap

Managers remain painfully ill-equipped while tech moves wearing silicon sneakers and powered by AI-smart hurricane tailwinds.



Winning: Deep Tech Appreciation

Cultivating a business perspective that deeply appreciates the role of technology to avoid being doomed.



Pillar 2: Strategic Missteps (The When & Why)

Treating IT as a Project

Failing: Installation Mindset

Treating IT merely as finite projects to install, configure, or integrate.



Winning: Evolution Mindset

Seizing tech as an opportunity to define and deploy entirely new ways of working.



Misjudging the Hype Cycle



3x Measured Scaling

Failing: The Burn vs. The Miss

Moving too early (cash burn) or too late (surrendering market dominance).



Winning: Chasing the Hype

Burning precious cash on unready tech or wallowing in multi-billion-dollar R&D without clear returns.



Poor Timing



Winning: Precision Entry

Timing entry perfectly to lock in competitive assets: network effects, scale, and switching costs.



Pillar 3: Execution Friction (The How)

Integration Gears



The Difficulty of Integration

Failing: Underestimating the Grind

Assuming that high budgets or elite mentors guarantee success, vastly underestimating how extraordinarily difficult transformation is.



Winning: Strategic Alignment

Acknowledging that even with significant backing, incompatible legacy systems and new tech cannot mesh without deep strategic alignment.



The Ultimate Fix: Building a Critical Thinking Muscle

Escaping the 72% failure rate requires an organization-wide capacity to rigorously evaluate new technologies and deeply understand their specific impacts on the firm and society. Moving from hype to sustainable integration is an ongoing discipline, not a one-time project.

Information Systems

Setting the Stage: Technology and the Modern Enterprise

Digital Exhaust is the New Oil

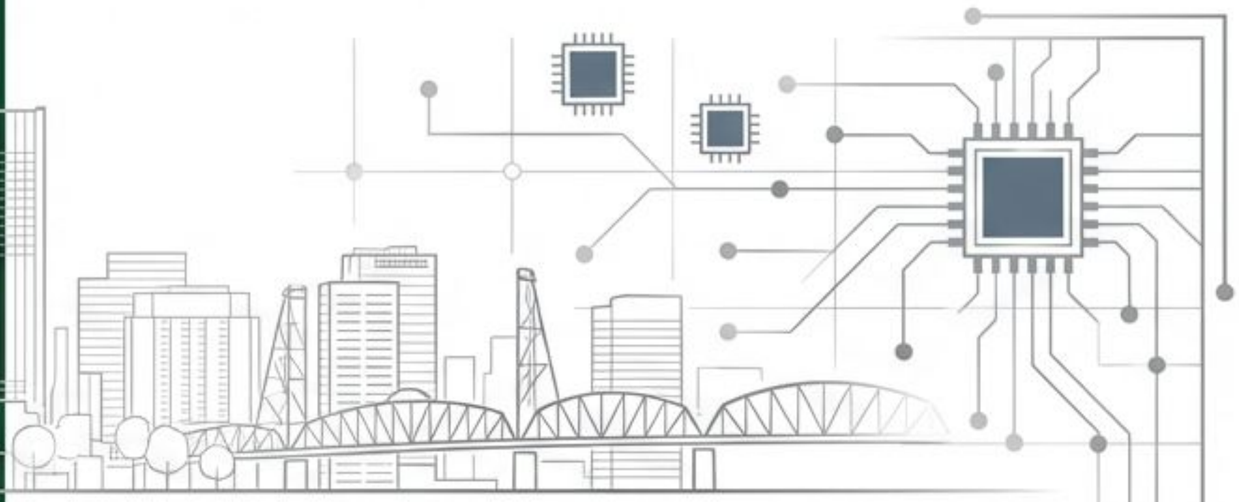
149 ZETTABYTES

Global data creation by 2024. 90% of this was created in the last 2 years.

Every click, scroll, and ping is raw material. It is useless until YOU give it meaning.

BA 325: Essentials of Information Literacy





The Silicon Forest Blueprint: Operating the Modern Enterprise

An Executive Briefing on Strategy, Scale, and Survival in the Digital Age.

Portland State University School of Business

Tech's Tectonic Shift



The Asset-Light Reality



Uber owns no cars. Airbnb owns no hotels. Alibaba owns no inventory. **The shift is already here.**


It’s Your Revolution
Technology democratizes disruption. You don’t need permission to build.

ANNOTATED ECOSYSTEM TIMELINE

The PC & Web

Bill Gates (Microsoft) & Larry/Sergey (Google) — Built on the personal and indexed computing.

Mobile & App Stores

Kevin Systrom (Instagram) — Scaled to 35M users with 13 employees in 18 months via App Stores.

Cloud & Open Source

Melanie Perkins (Canva) — Leveraged cloud computing to democratize graphic design to a $26B valuation.

Crowdfunding

Palmer Luckey (Oculus) — Raised $2.4M via Kickstarter, bypassing VC equity demands.



LOW-COST GLOBAL DISTRIBUTION: Near-zero marginal cost delivery allows digital products to reach massive markets instantly.

The Startup Idea

Do This

Not That

Build Prototypes

Test Product--Market Fit

Apply to Elite Accelerators (e.g., Y-Combinator)

The Golden Triad:

1. Find a problem.
2. Solve the problem.
3. Get paid to solve the problem.

Demand NDAs (Marks you as an amateur)

Obsess over Logo/Pitch Deck

Take "Dumb Money" (Cash without mentorship)


Geek Up
Moore's Law for People: Tech is embedding into every traditional management discipline.

FINANCE

Robo-Advisors & AI Trading:
Algorithms routinely beat major trading indices. Valuation requires technical due diligence.



MARKETING

The Algorithmic Funnel:
CRM, SEO, and the rise of the $21B Virtual Influencer economy.



HUMAN RESOURCES

The Data-Driven Workforce:
Algorithmic resume screening and precision talent poaching via LinkedIn.

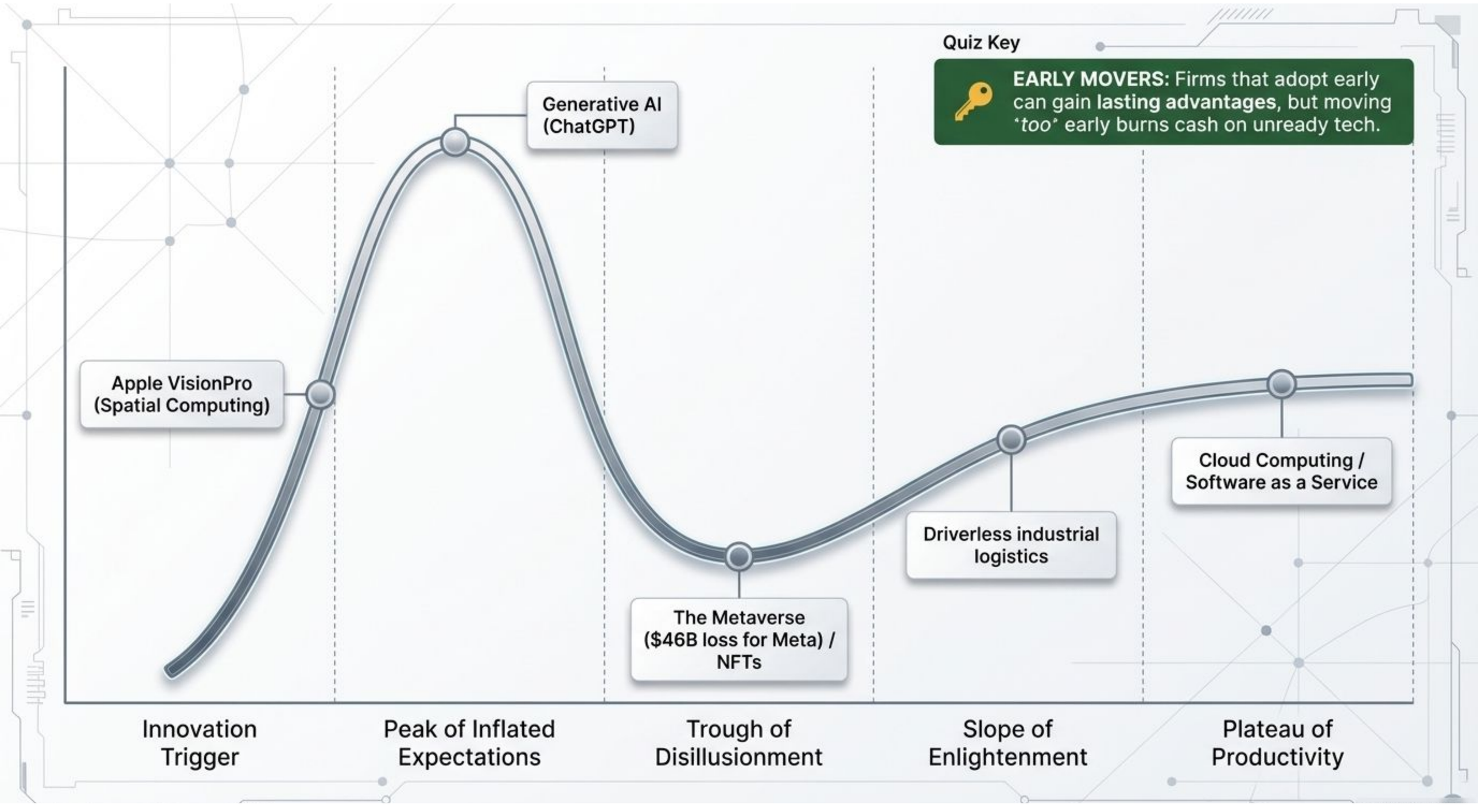


ACCOUNTING

Tech-Centric Audits:
Spreadsheets eliminated the math; AI is automating tax grunt work. Tech consulting now dwarfs traditional audit practices.




Hey Manager
Seeing beyond the noise to build durable advantages.


Quiz Key
EARLY MOVERS: Firms that adopt early can gain lasting advantages, but moving *too* early burns cash on unready tech.
Generative AI (ChatGPT)
Apple VisionPro (Spatial Computing)
The Metaverse ($46B loss for Meta) / NFTs
Driverless industrial logistics
Cloud Computing / Software as a Service
Innovation Trigger
Peak of Inflated Expectations
Trough of Disillusionment
Slope of Enlightenment
Plateau of Productivity

Operational Effectiveness



Doing similar things better. Necessary for survival, but not sufficient for dominance. Danger: Leads to pure price competition.

Strategic Positioning

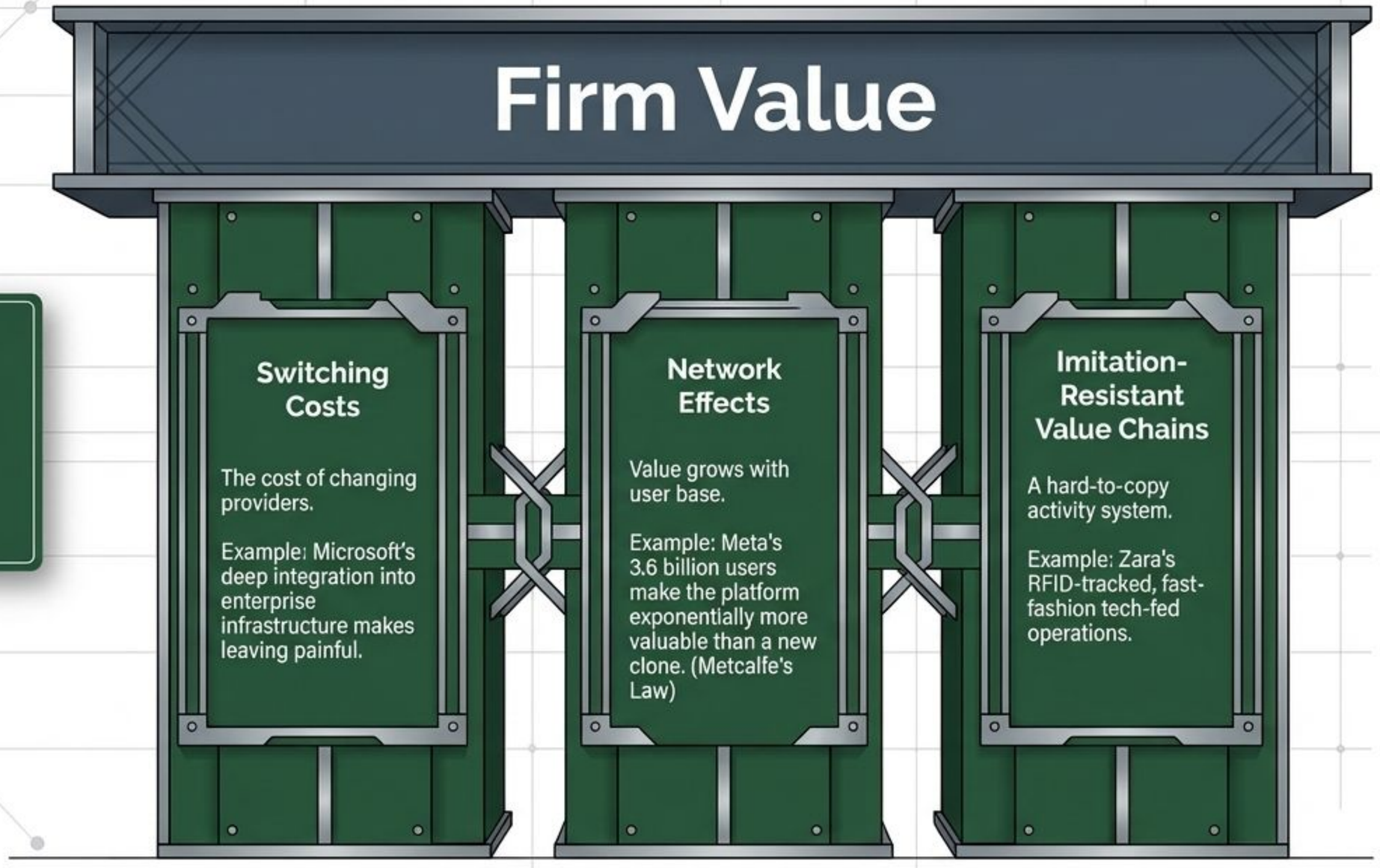


Choosing tradeoffs. Being meaningfully different. This is the true driver of profitability.



COMPETITIVE ADVANTAGE: The ultimate goal. The ability to outperform competitors and sustain profitability.

DIFFERENTIATION:
Building these pillars reduces pure price competition.



The Pages Ahead

Your strategic decoder ring for the modern enterprise.

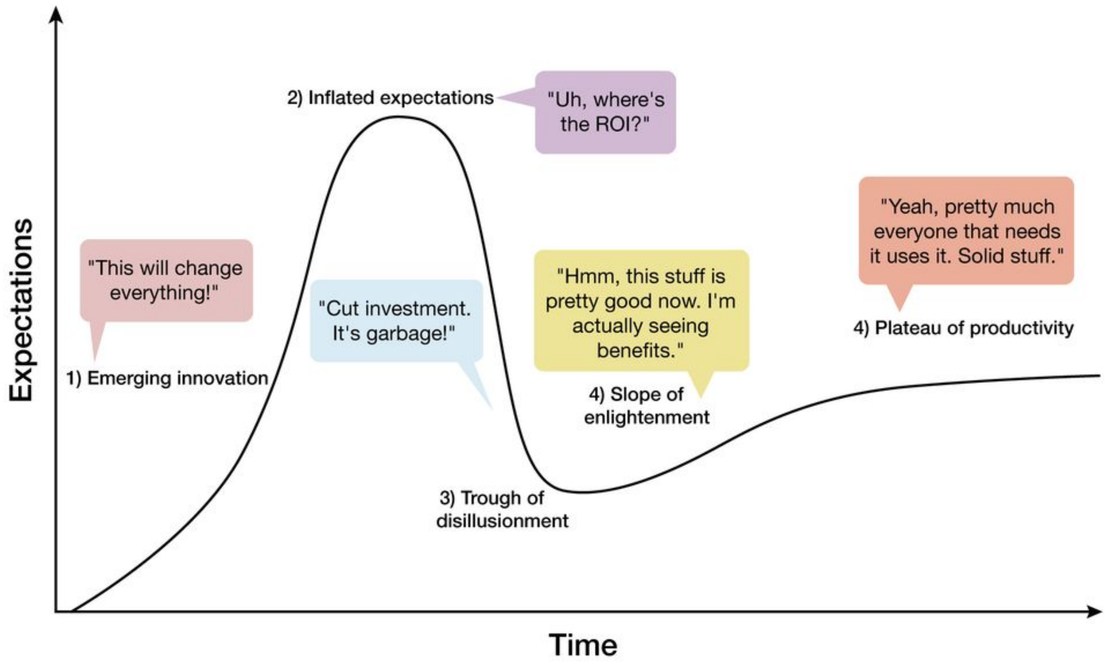
COMPANY	STRATEGY TOOLKIT MAPPING
Apple	Differentiation (Ecosystem, design, switching costs)
Netflix	Low-cost distribution (Global streaming at scale)
Meta	🔑 **NETWORK EFFECTS** Value grows exponentially with user base
Salesforce	APIs (Platform extensibility and integrations)
Amazon	Early mover advantage (Logistics and cloud dominance)

Reality Check: Firms rarely know outcomes in advance. Strategic advantage comes from acting early, learning fast, and adjusting **under uncertainty**.

Setting the Stage: The Future is Now!

- World’s largest companies own almost nothing
  - Uber: World's largest taxi service owns no cars.
  - Airbnb: World's largest accommodations provider owns no hotels or rental properties.
  - Facebook: World's most visited media destination creates no content of its own.
  - Temu, Etsy, Alibaba: Global retailers that own no product or inventory of their own.
- What is Driving This Change?
  - Moore’s Law: Microchips getting smaller and faster
  - Cloud Computing: The foundational driver that allows for immense scale without physical assets.
  - Artificial Intelligence (AI): The gas on the fire for rapid, disruptive change.

Navigating the Hype: Where are we Now?

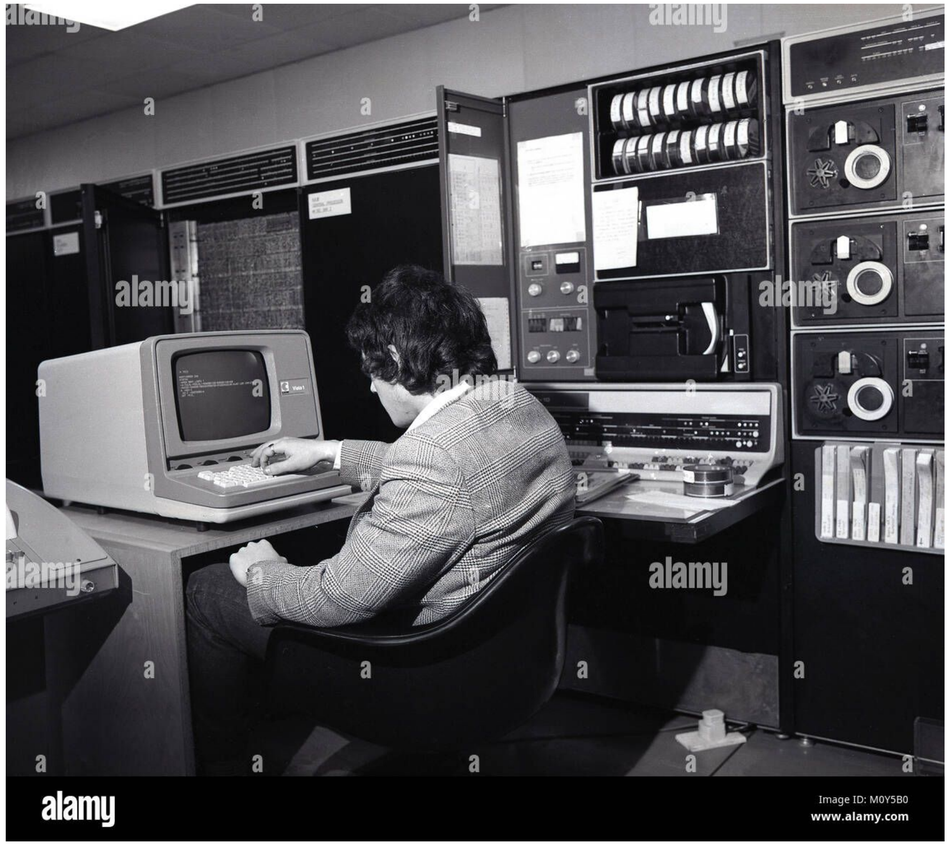


- The hype cycle is real
  - new tech follows a predictable path
- Textbook is looking through the eyes of the manager
- Your mission in BA 325
  - Critical thinking skills to separate hype from lasting business value
  - That's why we're learning foundational plumbing
  - Structure of data is a constant, no matter what the newest tech innovation is

Information Literacy

1970s (mid-late) → 1980s

- Banking:
  - Customers saw their passbooks replaced by computerized transaction printouts. Instead of a teller hand-writing balances, mainframes printed receipts.
- Airline tickets:
  - The SABRE system (by American Airlines and IBM, 1964) had gone mainstream. In the ’70s, a regular person might notice travel agents pulling up real-time flight data on terminals.
- Paychecks
  - Payroll increasingly run on computers. Instead of handwritten checks, employees started receiving standardized, computer-printed checks with itemized deductions.



Information Literacy (Chronology vs. Maturity)

How a Regular Person Saw Data in the 1970s

“Businesses have been collecting data since the 1970s, but collecting data is not the same as using it well. Half a century later, most organizations still struggle to turn data into better decisions.”

Data Maturity Model and Key Terms

Click into Sheets

Data Universe: Chronology and Function



Information Systems

Setting the Stage: Technology and the Modern Enterprise

Information Technology

The Infrastructure

- Hardware, servers, routers, APIs (Rocket Money, Paypal, Trading Platforms, etc.)
- Operating systems, databases, software
- Cloud storage, cybersecurity, networking
- Phone and communication

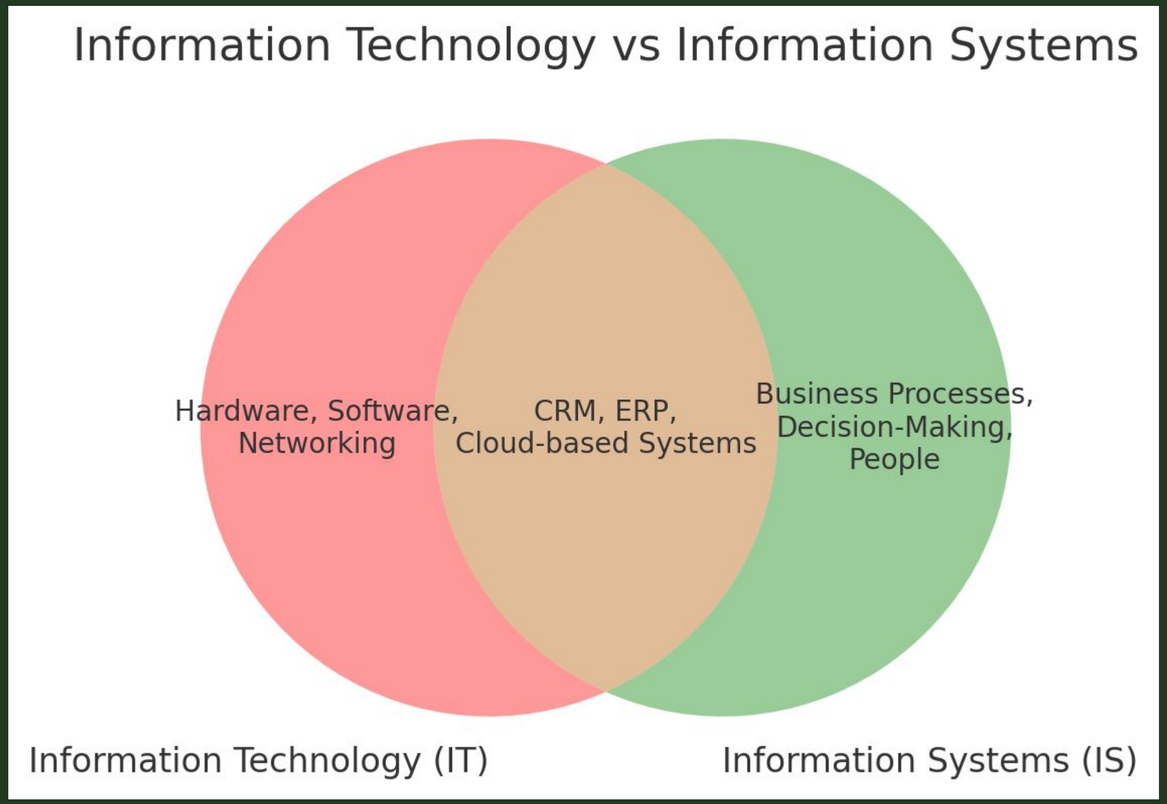
Information Systems

The Strategy and Application of Technology

- Business processes, how software is used
- Decision making
- People
- Achieving organizational goals

Modern Information Systems: Blurring the Lines

- Because modern information systems rely heavily on technology, IS and IT can be indistinguishable
- AWS, Google Cloud
  - Information Technology that creates value through connectivity of people (IS)
- CRM (Customer Relationship Management) Software
  - IT creates the tools that enable the process, but the technology is also the process itself
- Everything works in support of the movement and analysis of data



Warm-Up: What Do We Mean by “Data”?

- Take out a piece of paper and write your answers to the prompt displayed here →
- You have 3-5 minutes, write in complete sentences
- Be specific, give real-world examples

The term data is everywhere, in the news, in financial reports, in research, and inside every modern company.
In your own words, what is data, what is not data, give one real-world example from work, school, or daily life.

THE EVOLUTION OF DECISION-MAKING

THE HUNCH (GAMBLING)

- Gut feeling and intuition
- The **HiPPO** (Highest Paid Person's Opinion) dictates strategy
- Information isolated in functional silos
- Reactive scrambling

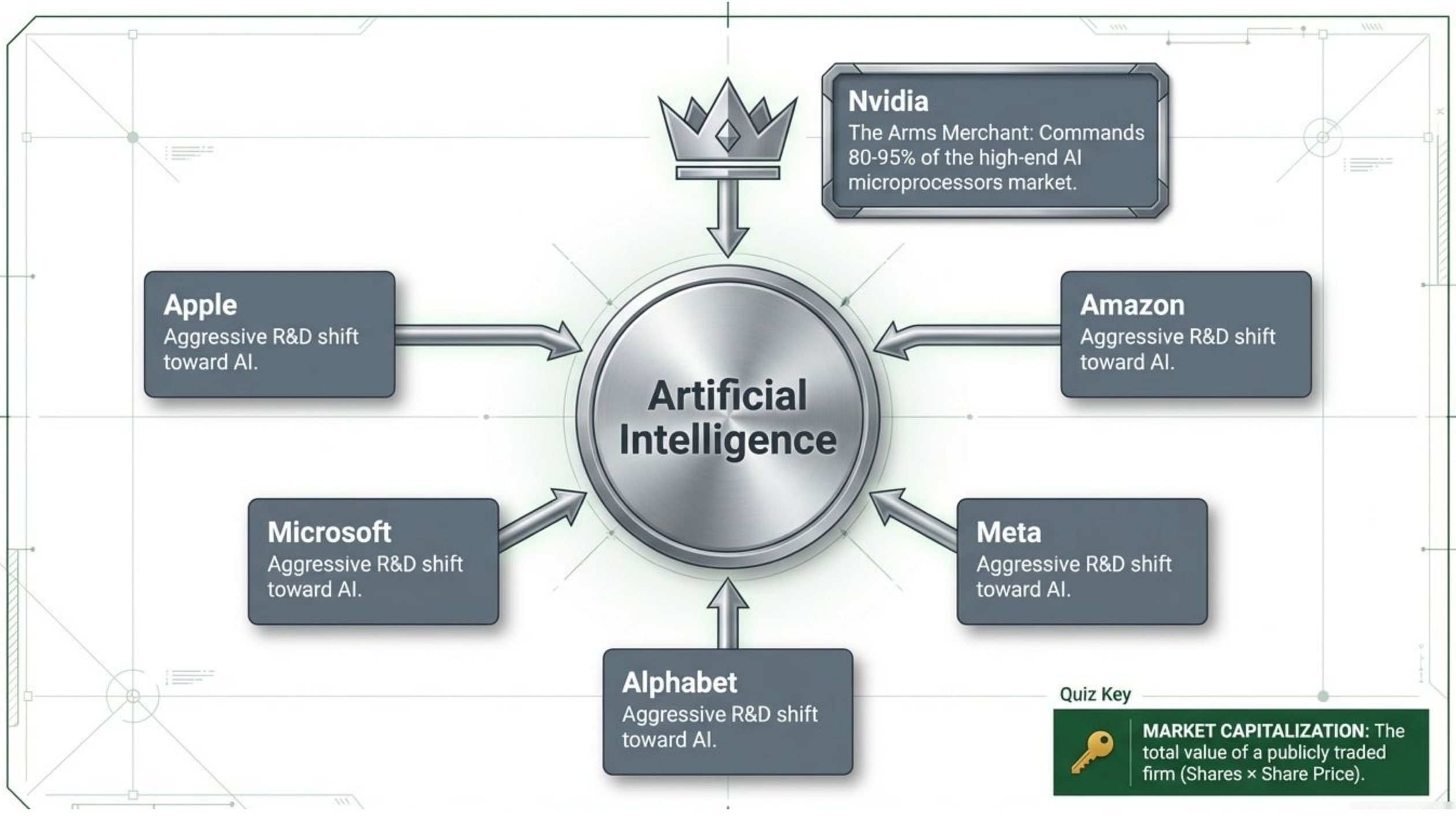
THE SCIENCE (MANAGING)

Data-backed and substantiated

Experiment-driven and iterative

Integrated, enterprise-wide systems

Predictive modeling


Nvidia
The Arms Merchant: Commands 80-95% of the high-end AI microprocessors market.
Apple
Aggressive R&D shift toward AI.
Amazon
Aggressive R&D shift toward AI.
Artificial Intelligence
Microsoft
Aggressive R&D shift toward AI.
Meta
Aggressive R&D shift toward AI.
Alphabet
Aggressive R&D shift toward AI.
Quiz Key
MARKET CAPITALIZATION: The total value of a publicly traded firm (Shares × Share Price).

The Promise

Accelerated COVID vaccine development.

GenAI coding assistance.

Automated legal & accounting grunt work.



The Peril

Algorithmic bias & hallucinations.

Copyright infringement of artists/writers.

Energy consumption.

Quiz Key | **BARRIER TO ENTRY: Massive capital requirements for Server Farms prevent new competitors from entering the AI infrastructure race.**